

Hamid Sourian, 59KG Greco-Roman Wrestler, Iran, Rio Olympic 2016
7 - 0 → Lost to a Fall (Pin)

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Outline: W06C

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k-Nearest Neighbor

IIII) Evaluation

Confusion Matrix → Multiclass

Fully Supervised Classification Evaluation

Technique (kNN, LR, SVM, …) does *not* matter …

Students (technique) learn from the course (labeled set) differently, yet all are evaluated/judged the same.



Statue of Lady Justice blindfolded and holding a balance and a sword, outside the Court of Final Appeal, Hong Kong

$$(o, c) \rightarrow f(o) \cong c$$

Lady Justice, or the goddess Justitia, an allegorical personification of justice and moral force in ancient Roman art known as Iustitia or Justitia. Her attributes are scales, a sword and blindfold.
https://en.wikipedia.org/wiki/Lady_Justice

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Fully Supervised Boolean Classification Evaluation

https://en.wikipedia.org/wiki/Precision_and_recall

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, Type II error	**True negative**	False omission rate (FOR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR−}}$
		False negative rate (FNR), Miss rate = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR−) = $\frac{\text{FNR}}{\text{TNR}}$	F_1 score = $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

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Fully Supervised Boolean Classification Evaluation

Contingency Table, Confusion Matrix

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, Type II error	**True negative**	False omission rate (FOR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR−}}$
		False negative rate (FNR), Miss rate = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR−) = $\frac{\text{FNR}}{\text{TNR}}$	F_1 score = $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

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Fully Supervised Boolean Classification Evaluation

Precision: How many of **the model's** catches (positive) are correct?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$

From the model's perspective ...

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Fully Supervised Boolean Classification Evaluation

Recall: How many of **the dataset's** positives are **remembered** by the model?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, Type II error			
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$			

From the **universe** perspective …

Outline: W06C

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k-Nearest Neighbor

IIII) Evaluation

Confusion Matrix → A Single Score from Both Views

F-Measure

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Fully Supervised Boolean Classification Evaluation

A Single Score: Weighted Harmonic Mean of Precision and Recall

$$F = \frac{1}{\alpha\frac{1}{P} + (1-\alpha)\frac{1}{R}} \quad \text{or} \left(\text{with } \beta^2 = \frac{1-\alpha}{\alpha}\right)$$

$$F_\beta = \frac{(\beta^2+1)PR}{\beta^2 P + R} \begin{cases} \beta > 1\text{: favors Recall} & \text{If } \beta \to +\infty, \text{ F} = \text{R} \\ \beta = 1\text{: } \quad 2\dfrac{\text{P} \times \text{R}}{\text{P} + \text{R}} & \\ 0 \le \beta < 1\text{: favors Precision} & \text{If } \beta \to 0.0, \text{ F} = \text{P} \end{cases}$$

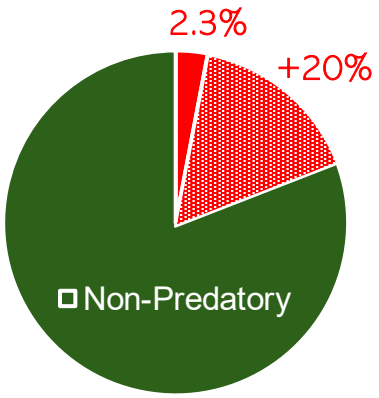
Fully Supervised Boolean Classification Evaluation

A Single Score: Weighted Harmonic Mean of Precision and Recall

Enhancing Online Grooming Detection via Backtranslation Augmentation*

https://hosseinfani.github.io/res/papers/2025_COLING_Enhancing_Online_Grooming_Detection_via_Backtranslation_Augmentation.pdf

original message	language: translation	backtranslation
'having it with minor'	French: *'l'avoir avec mineur'*	*'having sex with a minor'*
'i feel little aroused'	German: *'ich fühle mich ein wenig erregt'*	*'i'm feeling a little turned on'*
'you ever hooked up with anybody...?'	Chinese: '你有没有和网上的人交过?'	*'have you ever had sex with ...?'*
'like two guys doing each other?'	German: *'wie zwei typen, die es miteinander treiben?'*	*'like two guys having sex?'*



			Δf0.5			Δf1			Δf2		
			google	m2m100	nllb	google	m2m100	nllb	google	m2m100	nllb
gru	none		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	+fr		+8.76	+2.61	+2.65	+7.08	+2.84	+2.63	+5.80	+2.46	+3.35
	+fa		+1.68	+3.50	+4.05	+1.88	+3.64	+3.39	+2.43	+5.42	+3.04
	+de		+4.10	+3.81	+9.00	+3.76	+2.97	+6.39	+4.22	+2.92	+2.27
	+zh		-8.97	+1.55	+5.63	-5.87	+2.14	+4.27	-3.92	+4.89	+2.61
	+ca		+7.29	**+7.44**	+11.03	+6.73	+5.40	+8.04	+5.01	+2.67	+4.70
	+ps		9.57	-5.32	+13.60	+6.48	-3.79	+7.58	+3.04	+1.56	+0.52
	+is		+3.73	-2.24	+10.14	+4.04	-0.68	+6.98	+6.27	+2.25	+3.24
	+my		**+12.51**	-3.13	+9.34	+9.13	-1.66	+6.06	+4.63	+1.46	+2.00
	+fr+ca	Western Romance	+8.07	-4.40	**+15.29**	+7.02	-2.35	**+9.94**	+6.76	-0.81	+3.52
	+fa+ps	Iranic	-3.31	-6.71	+2.91	-1.88	-4.75	+2.62	+0.34	+0.78	+3.60
	+de+is	West Germanic	+11.84	+5.94	+9.43	**+9.39**	**+5.60**	+7.65	**+6.98**	**+6.63**	**+6.16**
	+zh+my	Sino-Tibetan	+8.83	-8.76	-3.15	+6.80	-6.33	-1.21	+4.55	-7.05	+0.66
	+fr+fa+de+zh	high-resource	+5.88	-1.36	+10.14	+4.62	-0.40	+7.63	+3.94	+3.01	+4.97
	+ca+ps+is+my	low-resource	+8.10	-1.38	+10.41	+6.61	-0.52	+5.26	+5.63	+2.64	-1.00
	all		+4.25	-0.32	+6.51	+4.30	+0.06	+5.20	+6.02	+2.16	+4.64

Outline: W06C

I) Task: Classification

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IIII) Evaluation

Confusion Matrix → Multiclass

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Fully Supervised Multiclass Classification Evaluation

		Truth	
		Like	Neutral
Model	Like	8	1
	Neutral	3	100

Facebook apologises for psychological experiments on users

The second most powerful executive at the company, Sheryl Sandberg, says experiments were 'poorly communicated'



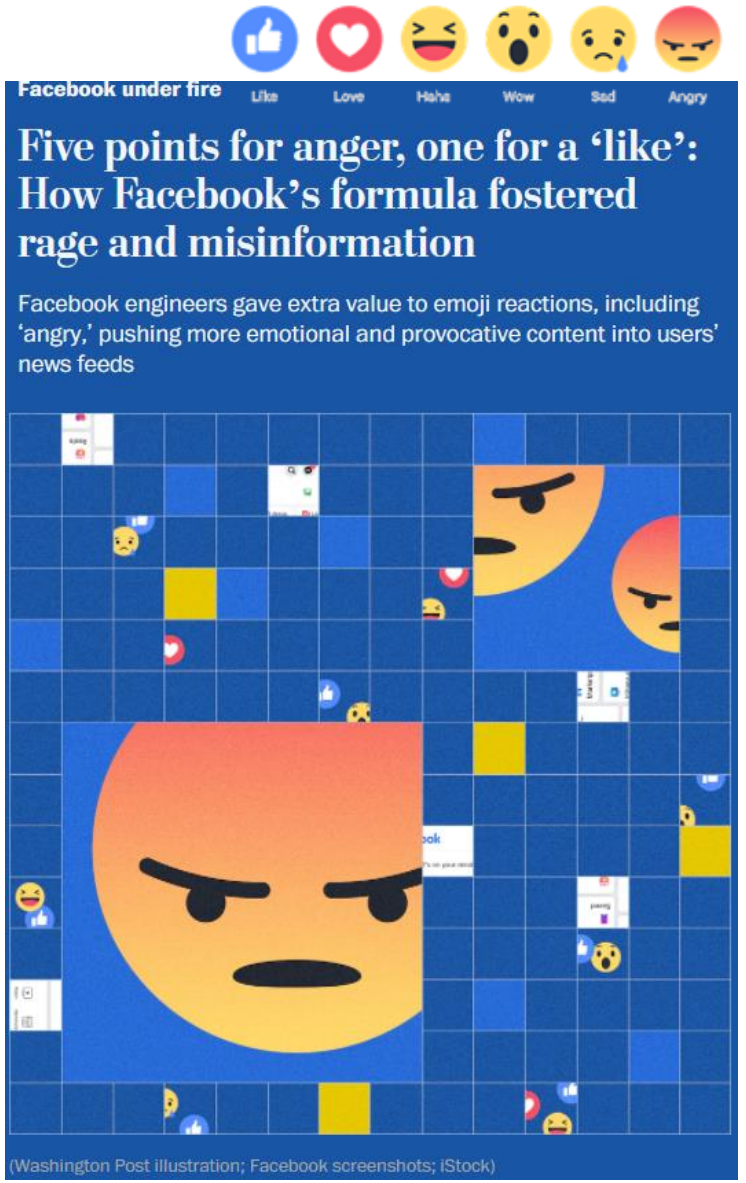
https://www.wsj.com/video/facebook-users-outraged-over-emotion-experiment/AFB5F0DF-3AB3-4E96-B2E1-1FA95D56A786



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Fully Supervised Multiclass Classification Evaluation

		Truth		
		Happy	Sad	Neutral
Model	Happy	8	10	1
	Sad	5	60	50
	Neutral	3	30	100



For 7 years, users begged for a **dislike** button. But Facebook **vehemently** resisted and feared a binary dislike button would introduce **cyberbullying, create a toxic/negative environment, or discourage people from posting entirely**.
https://www.nbcnews.com/tech/tech-news/emoji-reactions-cute-addition-facebook-became-headache-rcna3747
https://variety.com/2015/digital/news/facebook-dislike-sadness-sympathy-buttons-1201594213/
https://www.washingtonpost.com/technology/2021/10/26/facebook-angry-emoji-algorithm/

Fully Supervised Multiclass Classification Evaluation One-vs-Rest

		Truth		
		Happy	Sad	Neutral
Model	Happy	8	10	1
	Sad	5	60	50
	Neutral	3	30	100

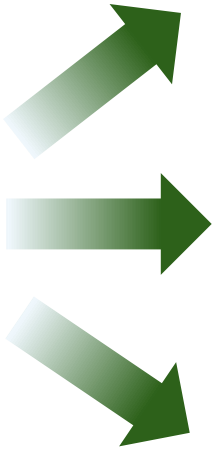
		Truth for Happy	
		Happy	Not Happy
Model	Happy		
	Not Happy		

		Truth for Sad	
		Sad	Not Sad
Model	Sad		
	Not Sad		

		Truth for Neutral	
		Neutral	Not Neutral
Model	Neutral		
	Not Neutral		

Fully Supervised Multiclass Classification Evaluation One-vs-Rest

		Truth		
		Happy	Sad	Neutral
Model	Happy	8	10	1
	Sad	5	60	50
	Neutral	3	30	100



		Truth for Happy	
		Happy	Not Happy
Model	Happy	8	11
	Not Happy	8	240

		Truth for Sad	
		Sad	Not Sad
Model	Sad	60	55
	Not Sad	40	112

		Truth for Neutral	
		Neutral	Not Neutral
Model	Neutral	100	33
	Not Neutral	51	83

Fully Supervised Multiclass Classification Evaluation One-vs-Rest

		Truth for Happy	
		Happy	Not Happy
Model	Happy	8	11
	Not Happy	8	240

$$P_{happy} = \frac{8}{8+11} = 0.42$$

$$R_{happy} = \frac{8}{8+8} = 0.5$$

		Truth for Sad	
		Sad	Not Sad
Model	Sad	60	55
	Not Sad	40	112

$$P_{sad} = \frac{60}{60+55} = 0.52$$

$$R_{sad} = \frac{60}{60+40} = 0.6$$

		Truth for Neutral	
		Neutral	Not Neutral
Model	Neutral	100	33
	Not Neutral	51	83

$$P_{neutral} = \frac{100}{100+33} = 0.75$$

$$R_{neutral} = \frac{100}{100+51} = 0.66$$

Fully Supervised Multiclass Classification Evaluation One-vs-Rest

		Truth		
		Happy	Sad	Neutral
Model	Happy	8	10	1
	Sad	5	60	50
	Neutral	3	30	100

$$P_{happy} = \frac{8}{8+11} = 0.42$$

$$P_{sad} = \frac{60}{60+55} = 0.52$$

$$P_{neutral} = \frac{100}{100+33} = 0.75$$

$$R_{happy} = \frac{8}{8+8} = 0.5 \qquad R_{sad} = \frac{60}{60+40} = 0.6 \qquad R_{neutral} = \frac{100}{100+51} = 0.66$$

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Fully Supervised Multiclass Classification Evaluation

One-vs-Rest: Macro Average $= \frac{1}{n}\sum_{i=1}^{n} \text{Metric}_n$

		Truth		
		Happy	Sad	Neutral
Model	Happy	8	10	1
	Sad	5	60	50
	Neutral	3	30	100

$$\frac{1}{3}[P_{happy} + P_{sad} + P_{neutral}]$$

$$P_{happy} = \frac{8}{8+11} = 0.42$$

$$P_{sad} = \frac{60}{60+55} = 0.52$$

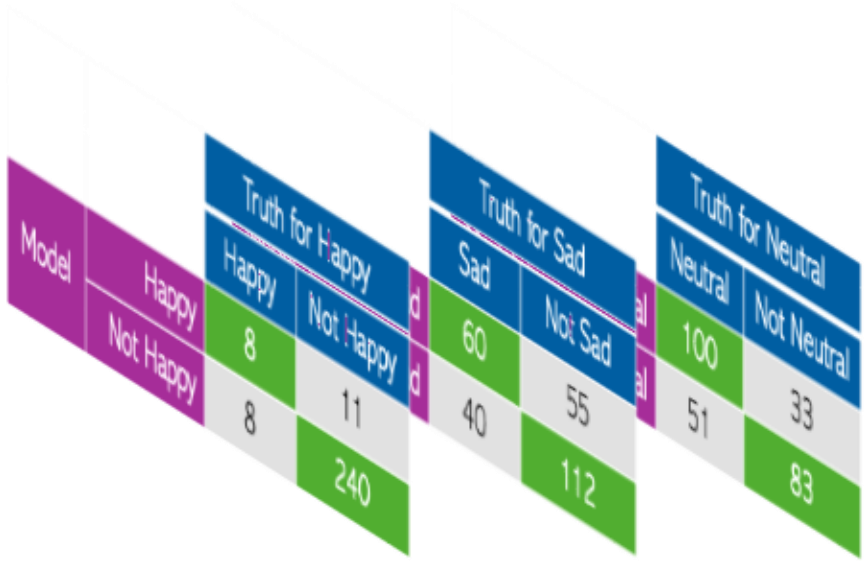
$$P_{neutral} = \frac{100}{100+33} = 0.75$$

$$R_{happy} = \frac{8}{8+8} = 0.5 \qquad R_{sad} = \frac{60}{60+40} = 0.6 \qquad R_{neutral} = \frac{100}{100+51} = 0.66$$

$$\frac{1}{3}[R_{happy} + R_{sad} + R_{neutral}]$$

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Fully Supervised Multiclass Classification Evaluation
One-vs-Rest: Micro Averaging



Pooling

$$(o, c) \rightarrow f(o) \cong c$$

Model		True	Not True
	True	168	99
	Not True	99	?

$$P_{True} = \frac{168}{168+99}$$

$$R_{True} = \frac{168}{168+99}$$

They become the same! Always for single-label multiclass classification. Why?

Precision and recall, regardless of classes, become …

Fully Supervised Multiclass Classification Evaluation

One-vs-Rest: Macro vs. Micro Averaging

Micro is dominated by the more frequent class.

Micro better reflects the performance on smaller classes.

Micro more appropriate when all classes are equally important.

Outline: W06C

I) Task: Classification

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IIII) Evaluation

Metrics, Methodology: How to Run Evaluation?

(1 Final) vs. (2 Midterms + 1 Final)

Evaluation Methodology: Protocol to Count Inferences

Contingency Table, Confusion Matrix

		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error
	Predicted condition negative	**False negative**, Type II error	**True negative**

What would be the observations (questions)

On which the model (student) make inferences (answers)

About their labels (true answers)…

Evaluation Methodology: Protocol to Count Inferences

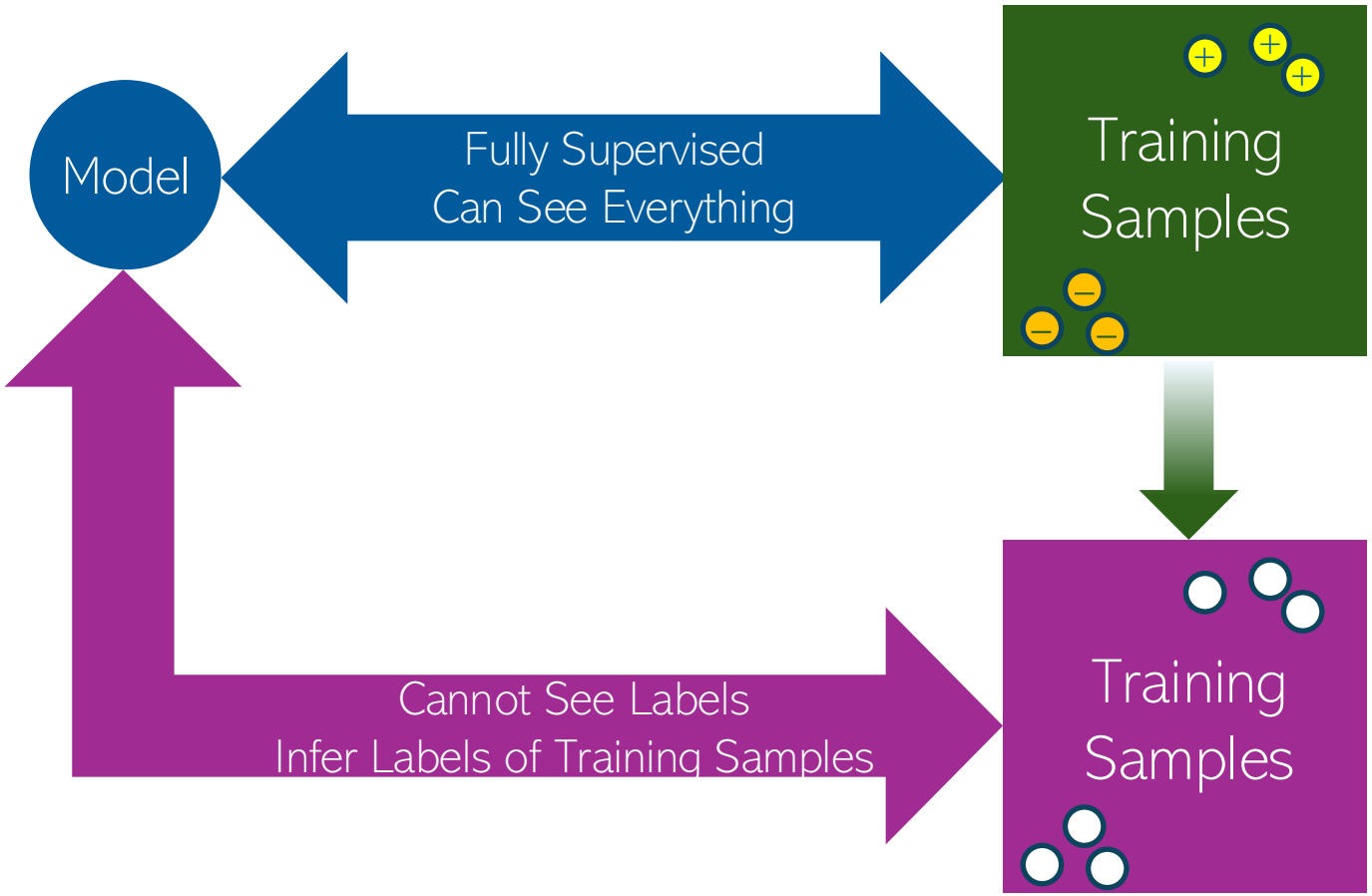


Students have the questions and answers in the slides.
Now, evaluate their performance on what questions?

Evaluation Methodology: Protocol to Count Inferences

Evaluate on the **same training set**

Simple, but time consuming (training set may be large)

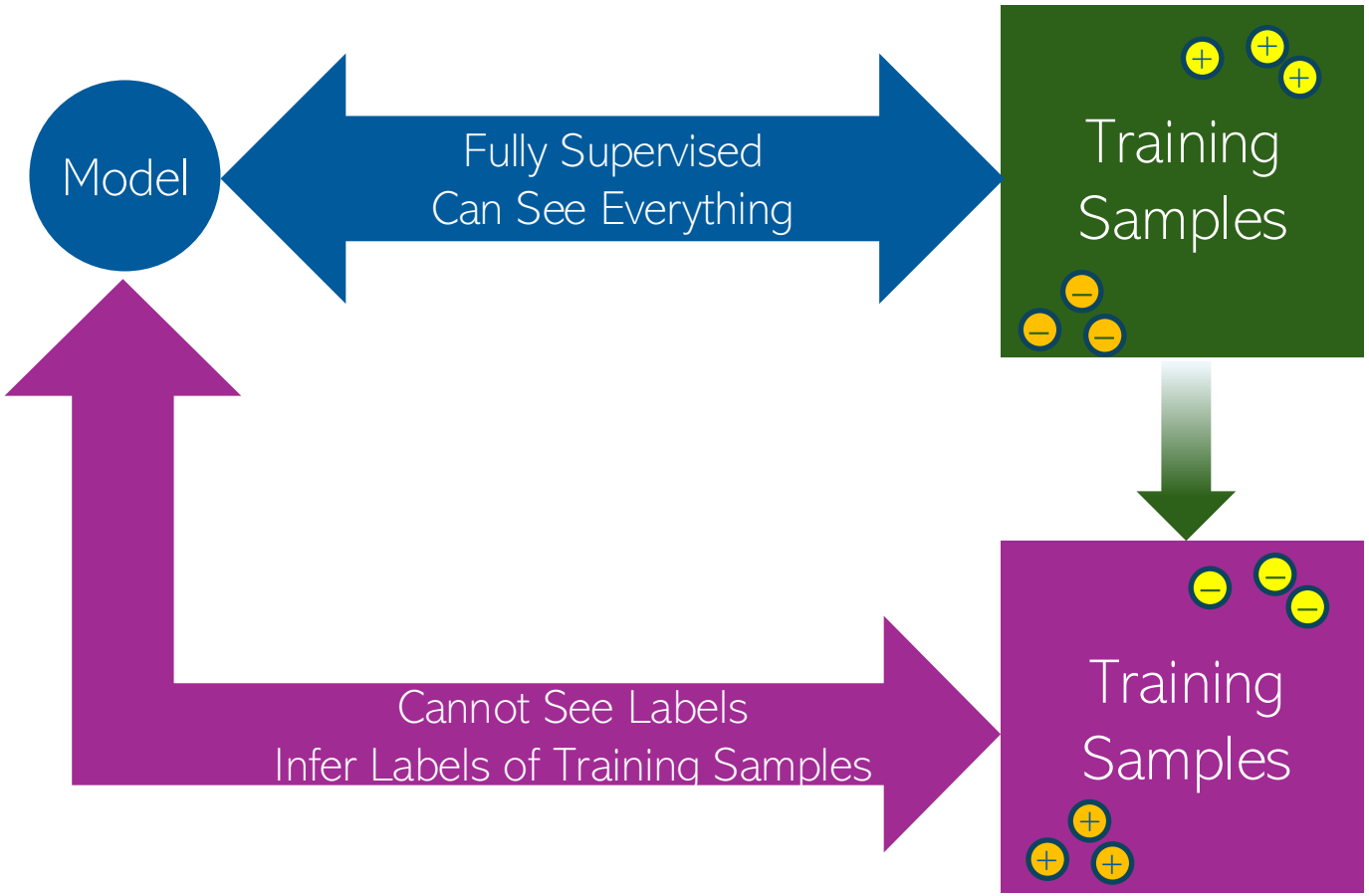


		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error
	Predicted condition negative	**False negative**, Type II error	**True negative**

Evaluation Methodology: Protocol to Count Inferences

Evaluate on the **same training set**

Simple, but time consuming (training set may be large)



Underfitting

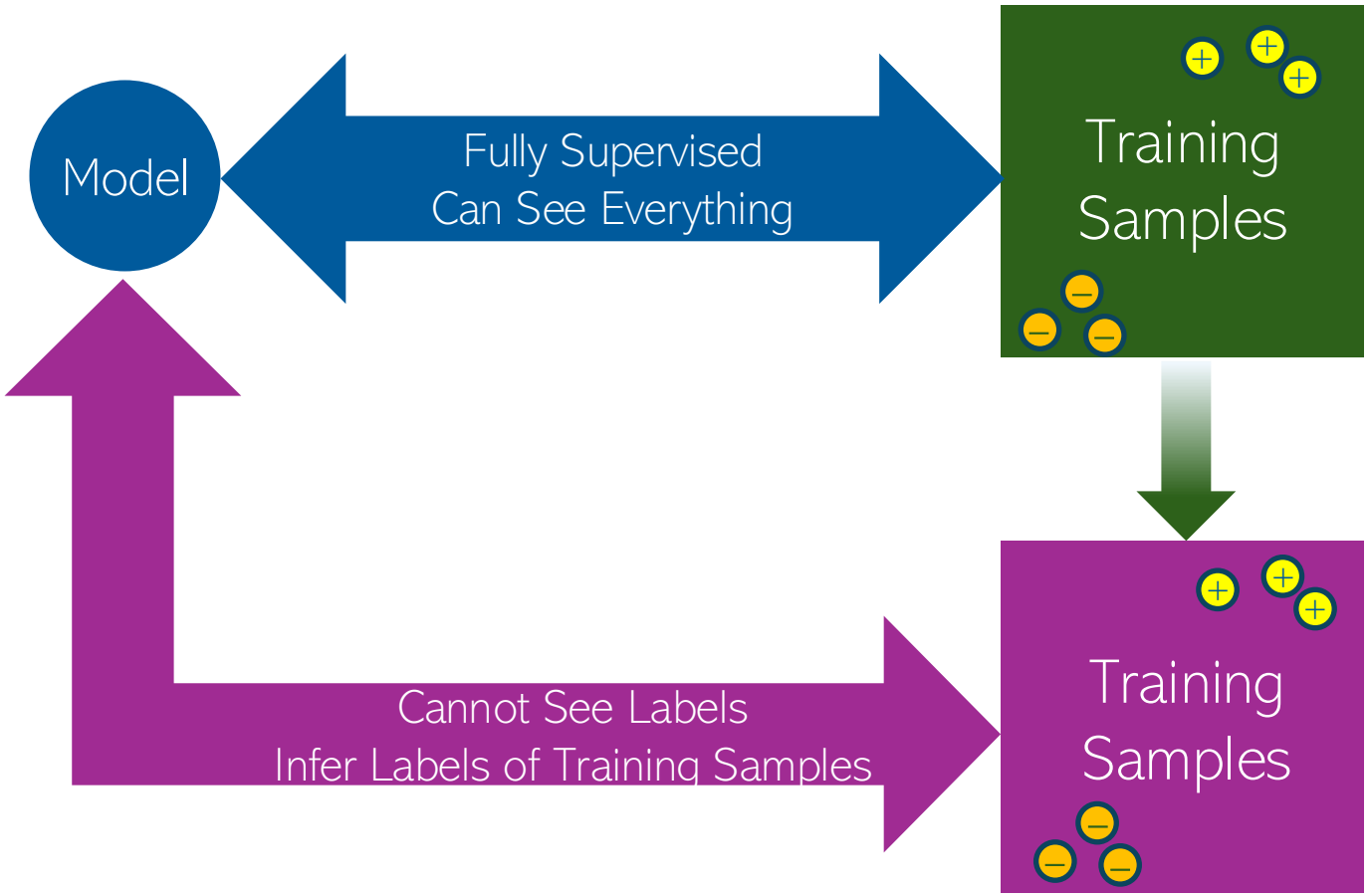
The model cannot learn!

		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	0 **True positive**	3 **False positive**, Type I error
	Predicted condition negative	3 **False negative**, Type II error	0 **True negative**

Evaluation Methodology: Protocol to Count Inferences

Evaluate on the **same training set**

Simple, but time consuming (training set may be large)



The model can learn.

An Oracle? Of course not!

It already knew the labels. No magic ...

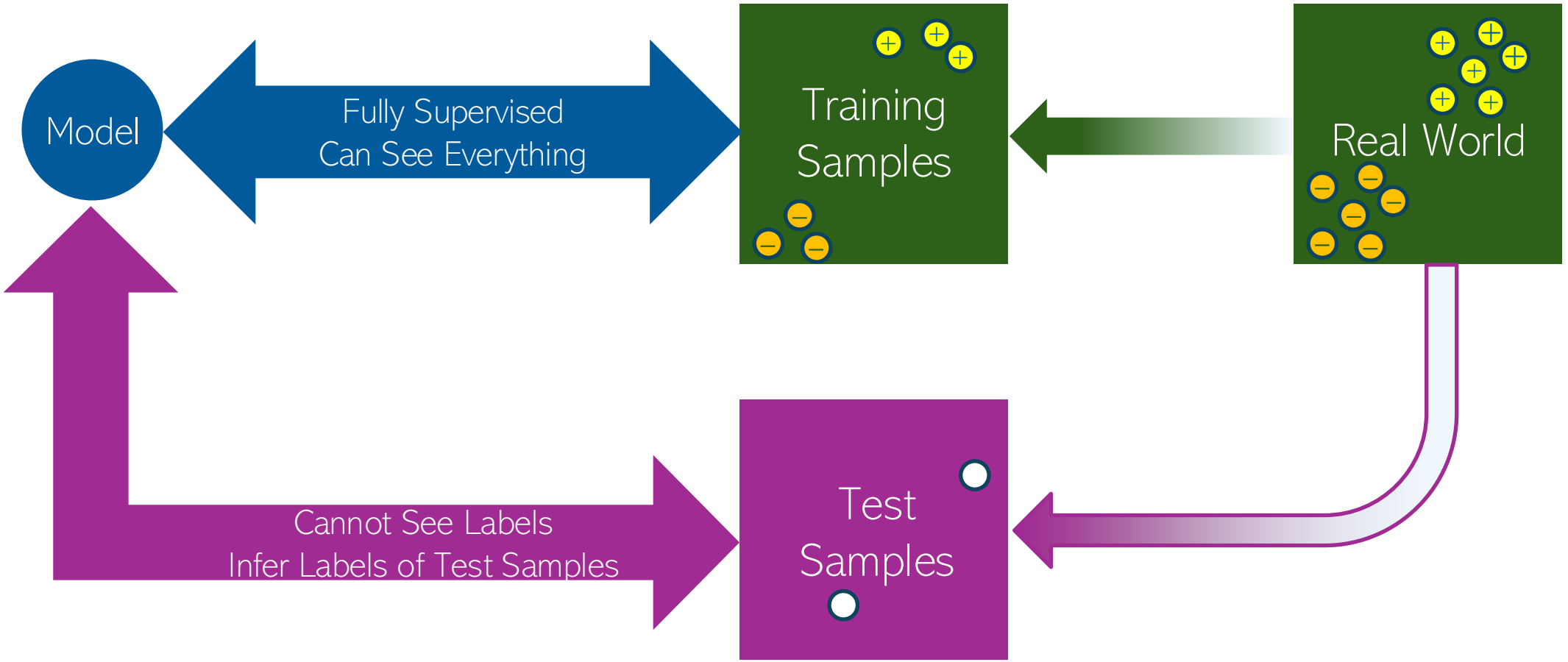
		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	3 **True positive**	0 **False positive**, Type I error
	Predicted condition negative	0 **False negative**, Type II error	3 **True negative**



Evaluation Methodology: Protocol to Count Inferences

Evaluate on a different **unseen samples: test set**

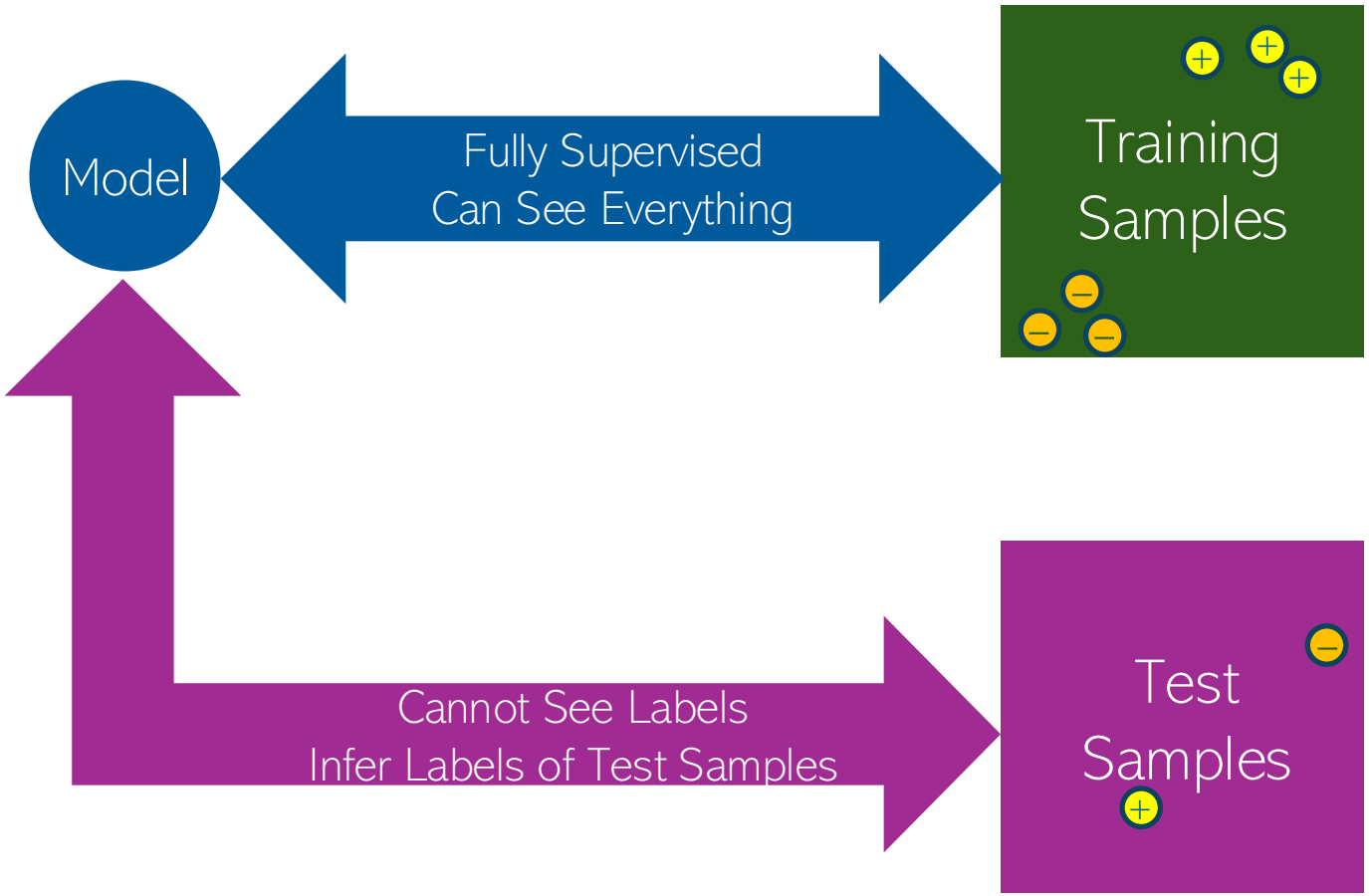
Simple, also fast (test set is commonly small), but very challenging



Evaluation Methodology: Protocol to Count Inferences

Evaluate on a different **unseen samples: test set**

Simple, also fast (test set is commonly small), but very challenging



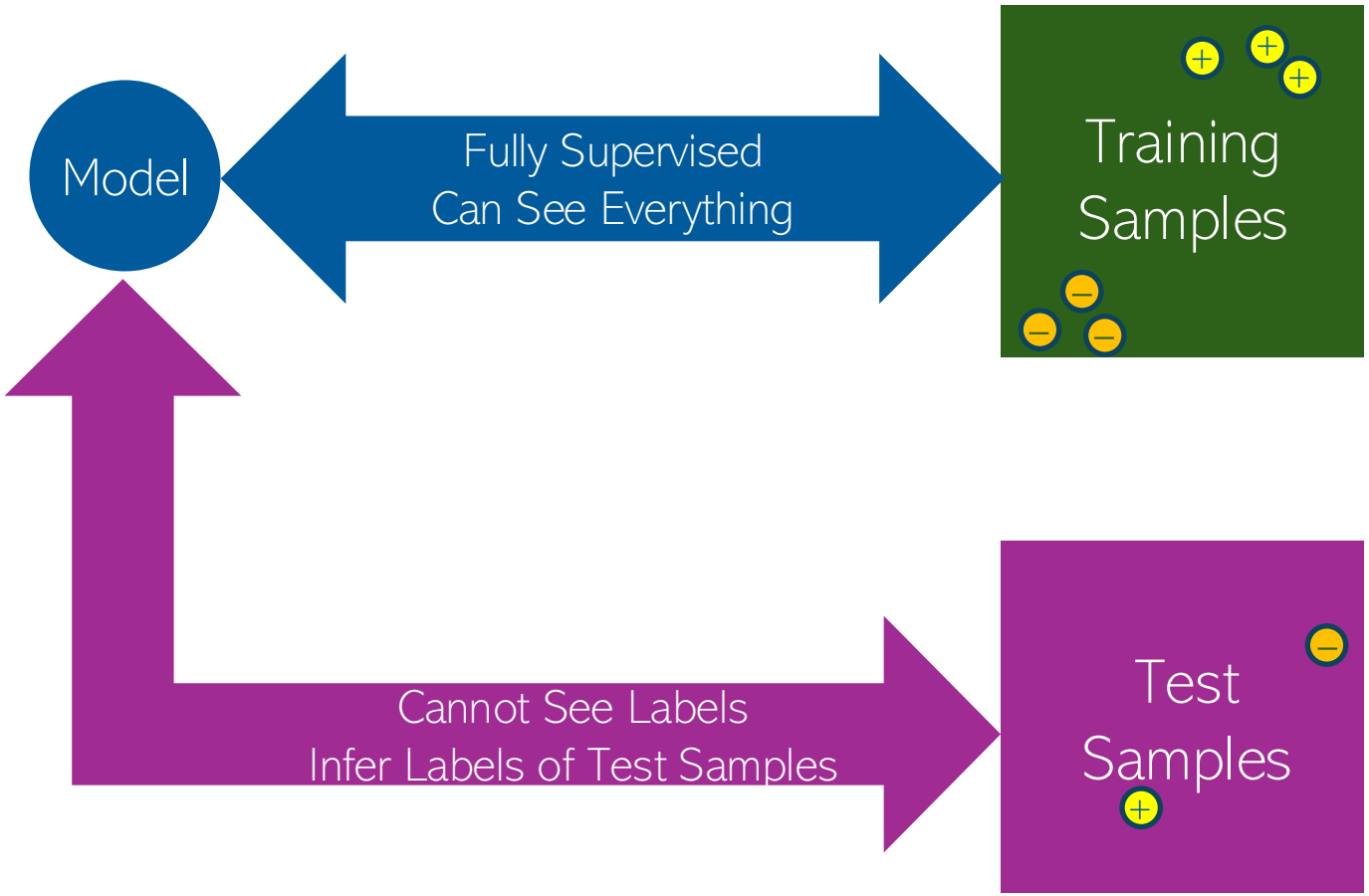
Poor Performance. Why?

		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	0 **True positive**	1 **False positive**, Type I error
	Predicted condition negative	1 **False negative**, Type II error	0 **True negative**

Evaluation Methodology: Protocol to Count Inferences

Evaluate on a different **unseen samples: test set**

Simple, also fast (test set is commonly small), but very challenging



		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	0 **True positive**	3 **False positive**, Type I error
	Predicted condition negative	3 **False negative**, Type II error	0 **True negative**

Underfitting Model. Expected.

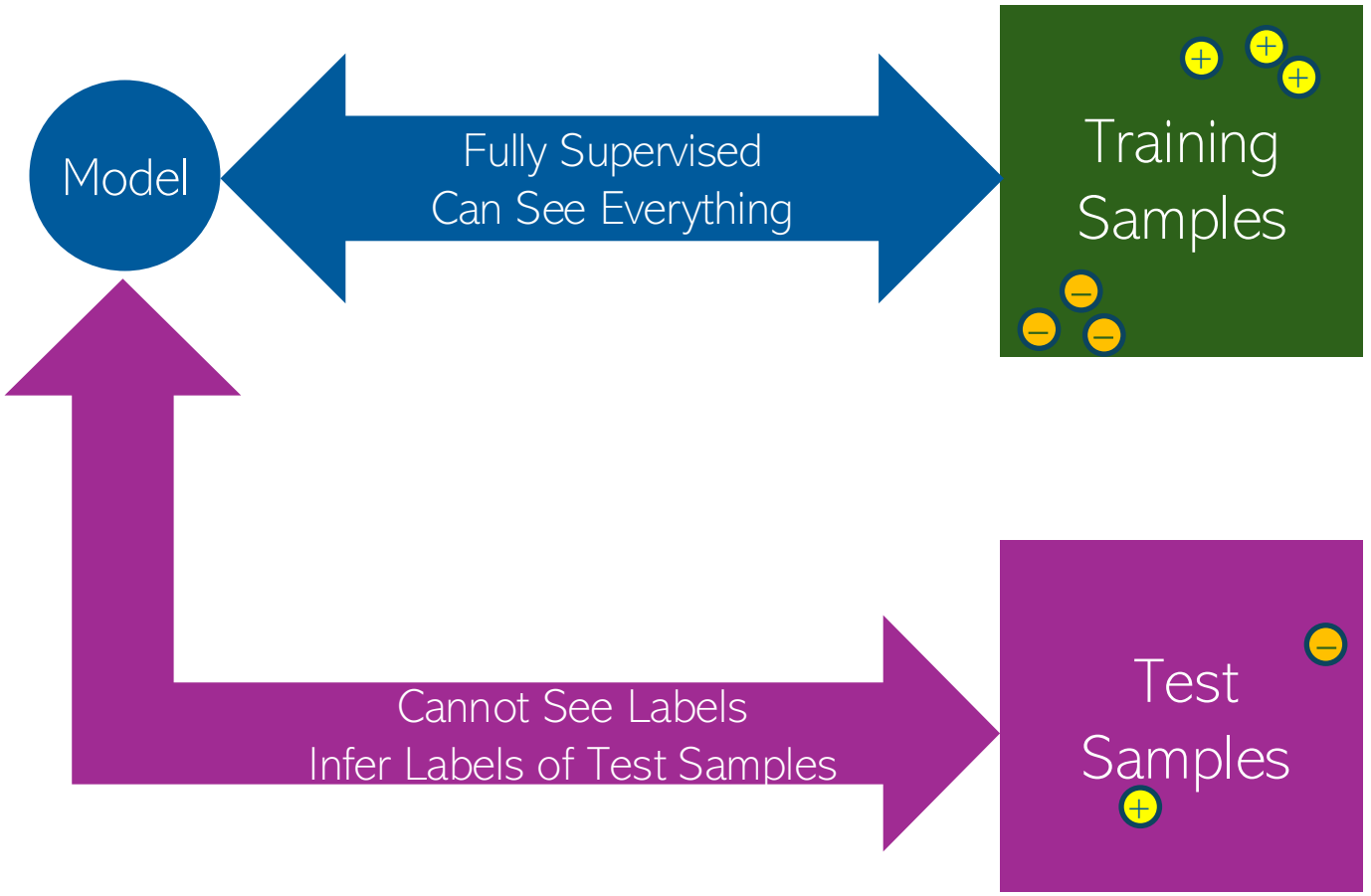
We should fix the model.

		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	0 **True positive**	1 **False positive**, Type I error
	Predicted condition negative	1 **False negative**, Type II error	0 **True negative**

Evaluation Methodology: Protocol to Count Inferences

Evaluate on a different **unseen samples: test set**

Simple, also fast (test set is commonly small), but very challenging



	Total population	True condition: Condition positive	True condition: Condition negative
Predicted condition	Predicted condition positive	3 **True positive**	0 **False positive**, Type I error
Predicted condition	Predicted condition negative	0 **False negative**, Type II error	3 **True negative**

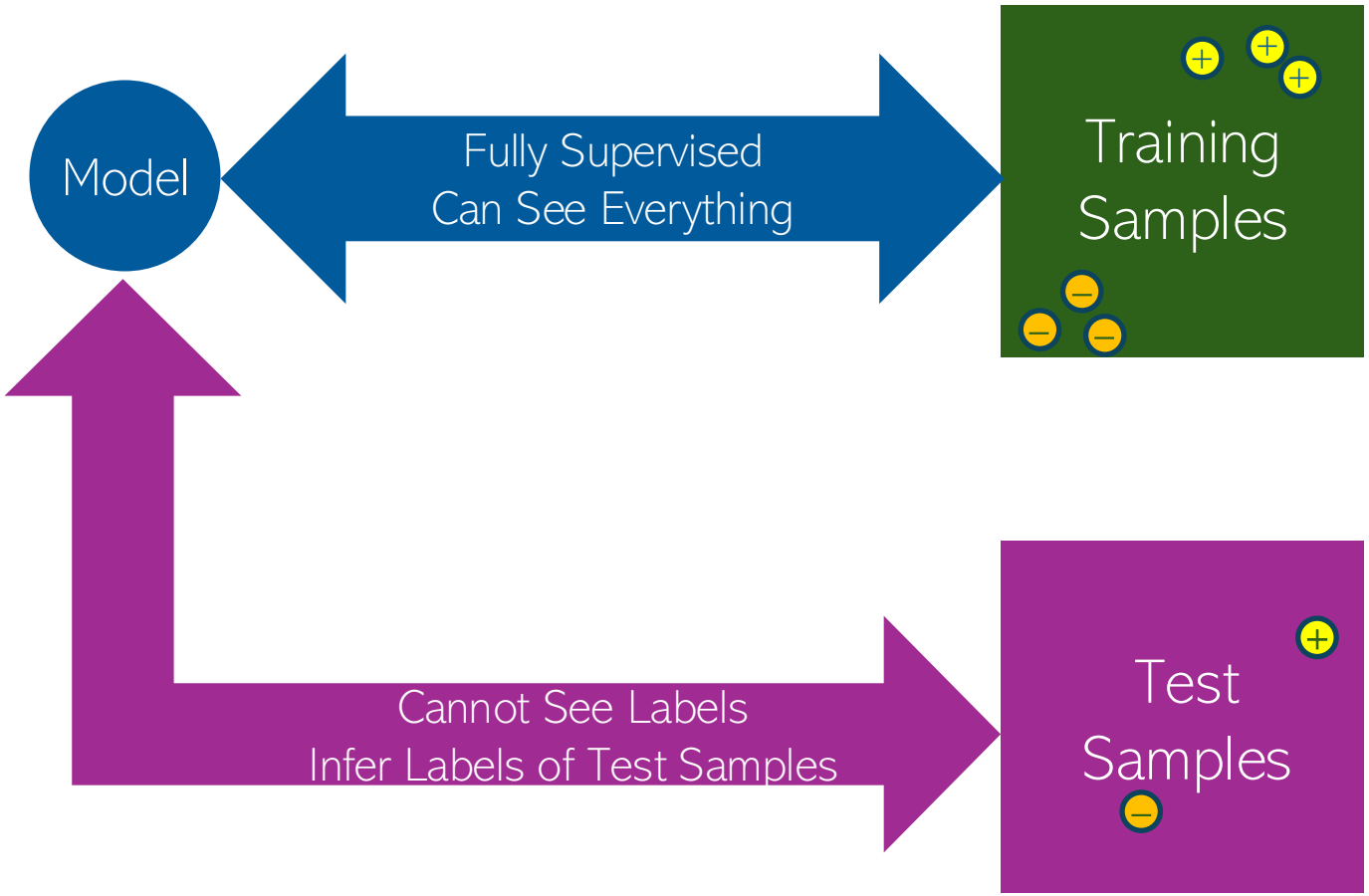
Overfitting to Training Set.
The model cannot generalize.

	Total population	True condition: Condition positive	True condition: Condition negative
Predicted condition	Predicted condition positive	0 **True positive**	1 **False positive**, Type I error
Predicted condition	Predicted condition negative	1 **False negative**, Type II error	0 **True negative**

Evaluation Methodology: Protocol to Count Inferences

Evaluate on a different **unseen samples: test set**

Simple, also fast (test set is commonly small), but very challenging



An Oracle.

No need to evaluate on train set.

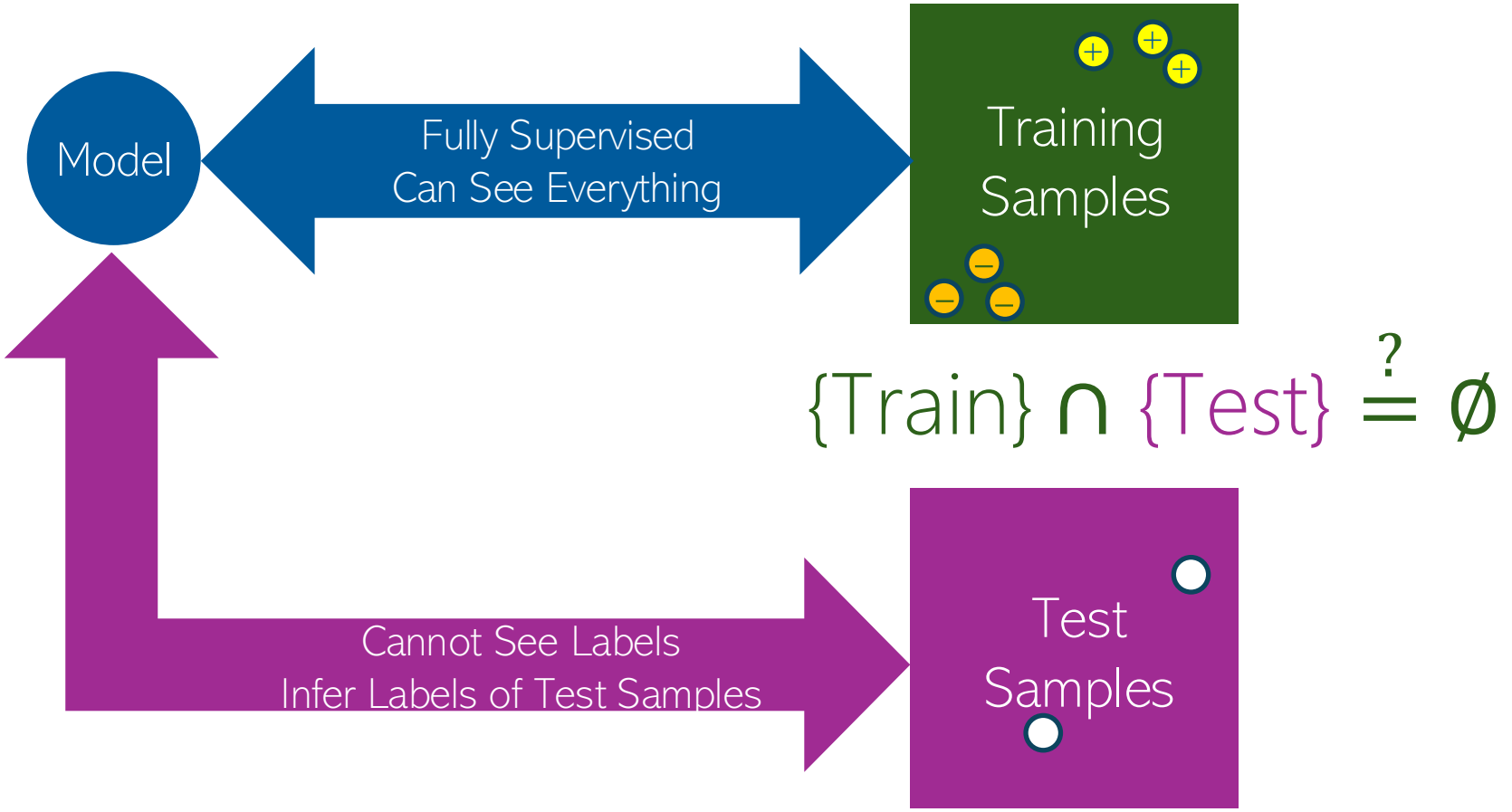
Rarely happens!

		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	1 **True positive**	0 **False positive**, Type I error
	Predicted condition negative	0 **False negative**, Type II error	1 **True negative**

Evaluation Methodology: Protocol to Count Inferences

Should training and test sets be **disjoint**, or an **overlap** acceptable?

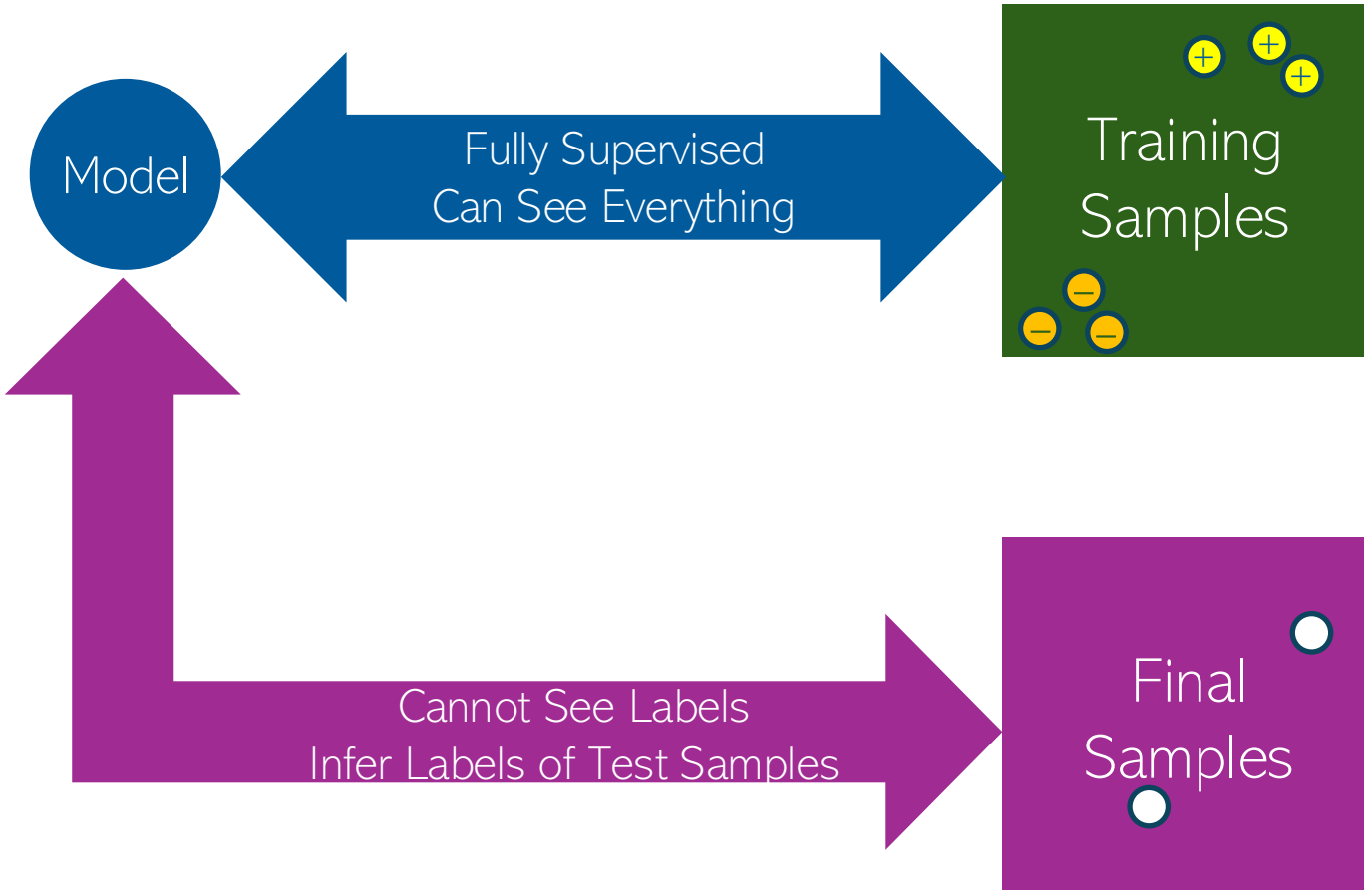
Think of students in an exam. Should all questions be unseen before?



Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a **single test set** enough to decide on a model?

*Think of students in **only one exam** (final exam). Is it sufficient to pass/fail?*

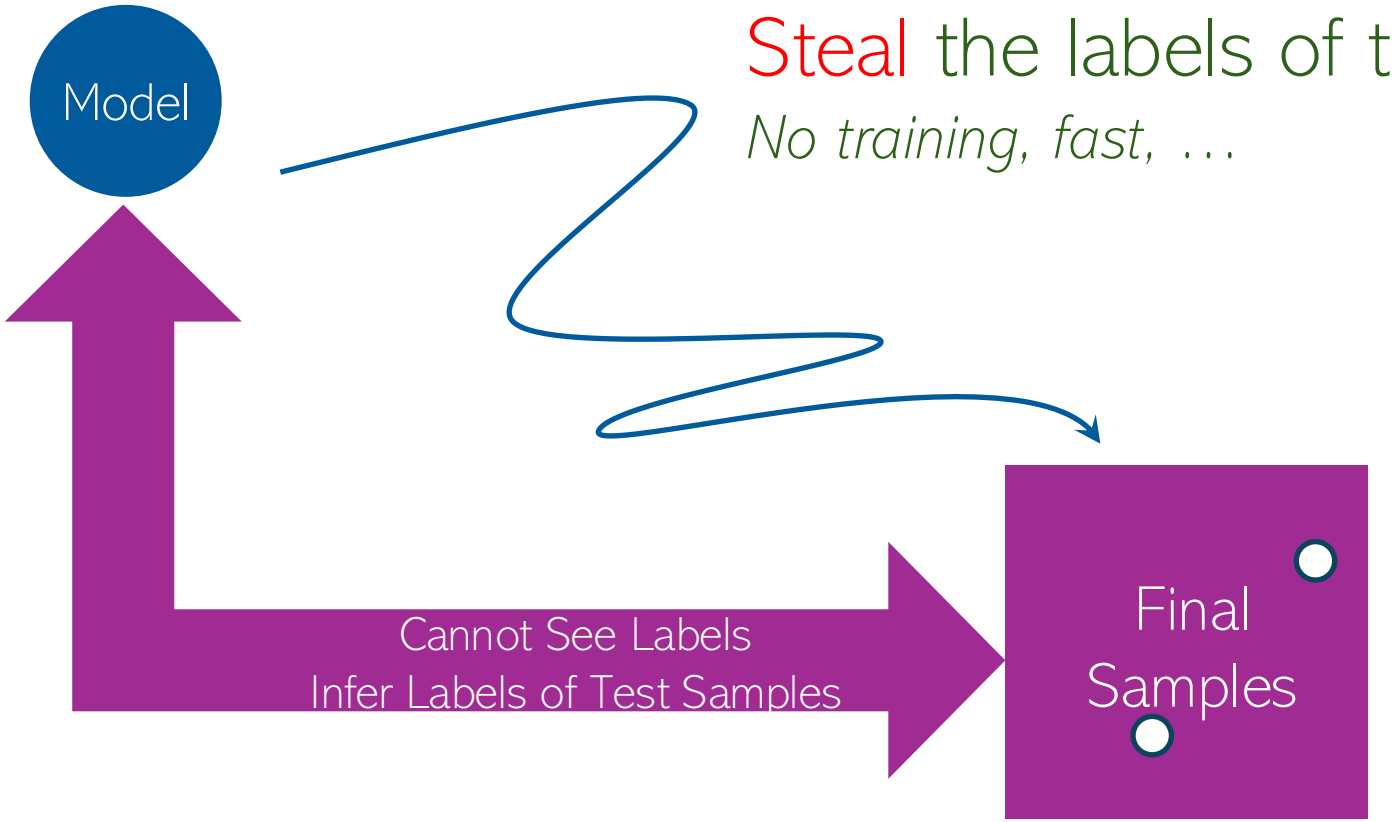


Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a **single test set** enough to decide on a model?

*Think of students in **only one exam** (final exam). Is it sufficient to pass/fail?*

Yes.







Mongrel Jedi

PetFinder.my Contest: 1st Place Winner Disqualified

307

Posted in petfinder-adoption-prediction a year ago

Dear Participants,

We would like to announce that the 1st Place Team, Bestpetting has been disqualified from the contest for cheating. The Kaggle Grandmaster cheater has also been permanently banned on this platform as the evidence points towards him being the key party behind this fraudulent activity.

Here is what the Bestpetting team did in the PetFinder.my contest:

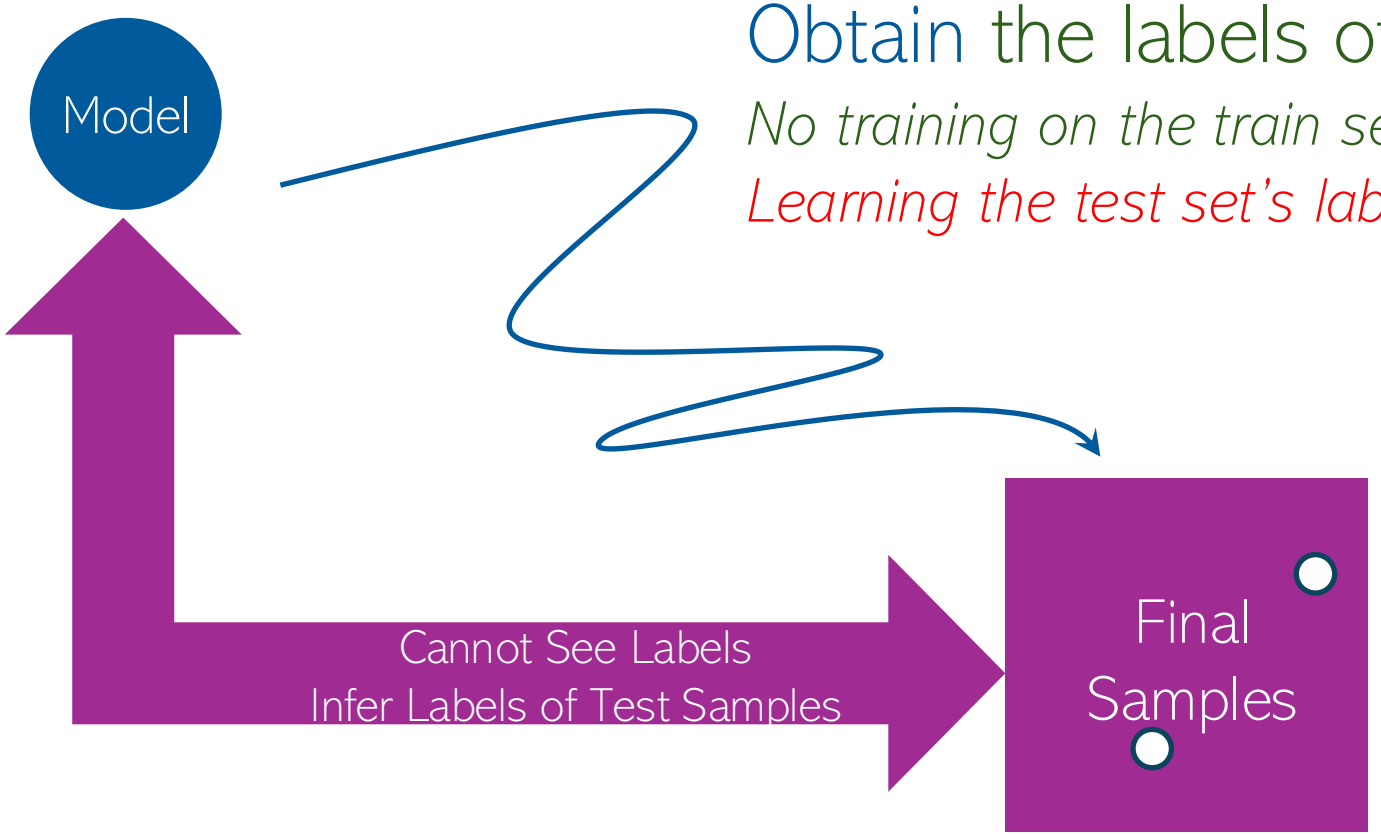
- They fraudulently obtained adoption speed answers for the private test data (possibly by scraping our website)

Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a **single test set** enough to decide on a model?

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Yes.



Obtain the labels of the test set!

No training on the train set but on evaluation results!

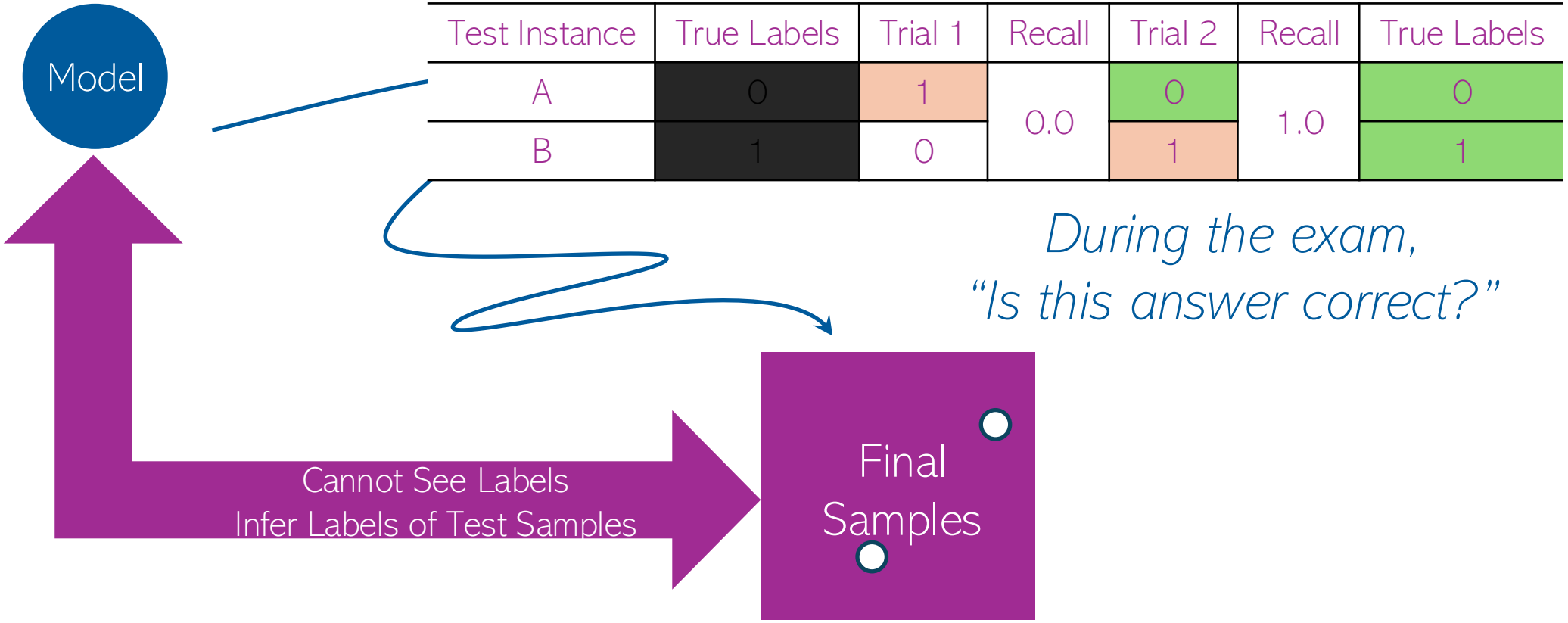
Learning the test set's labels.

Evaluation Methodology: Protocol to Count Inferences

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*Think of students in **only one exam** (final exam). Is it sufficient to pass/fail?*

Yes.

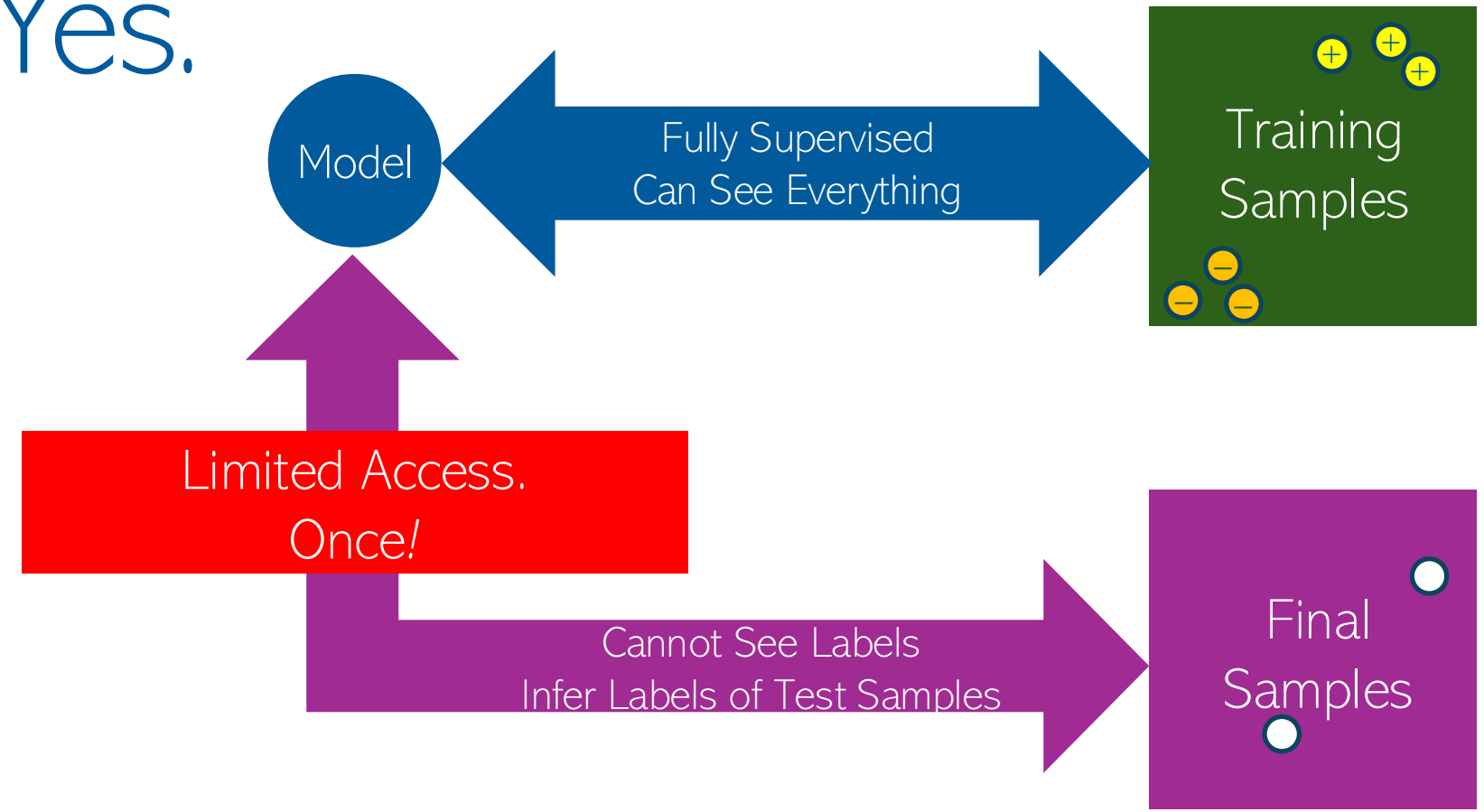


Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a **single test set** enough to decide on a model?

*Think of students in **only one exam** (final exam). Is it sufficient to pass/fail?*

Yes.



Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a **single test set** enough to decide on a model?

*Think of students in **only one exam** (final exam). Is it sufficient to pass/fail?*

No.

What if the student was not feeling well.
What if the test set favors students good at math but discount students good at critical thinking.
What if the students have stolen the keys.
What if ...